Hydrocortisone

Mechanism of action

- Hydrocortisone is a corticosteroid with anti-inflammatory and immunosuppressant actions. It inhibits the production of inflammatory mediators, including prostaglandins and leukotrienes, resulting in a reduction in the inflammatory and immune response.

Authorised practice levels

- Paramedics:
  - At risk of adrenal crisis or suspected to have adrenal crisis.
  - Severe or immediately life-threatening asthma.
  - Severe COPD or imminent respiratory arrest from COPD.
- ICPs and CCPs: All indications.

Indications

- ✓ Adults with angioedema occurring during inter-hospital transfer for stroke clot retrieval.
- ✓ At risk of adrenal crisis or suspected to have adrenal crisis.
- ✓ Severe or immediately life-threatening asthma.
- ✓ Severe COPD or imminent respiratory arrest from COPD.

Contraindications

- ✕ Known severe allergy.

Cautions

- ⚠ None.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated during pregnancy. However, there is significant clinical experience with steroids, they appear to be safe and hydrocortisone should be administered if indicated.

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- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- 100 mg for an adult.
- See the paediatric drug dose tables for a child.

Administration

- Press down on the plastic cap to force the diluent into the vial.
- Mix the solution by turning the vial upside down several times. Do not shake the vial.
- Remove the central plastic cap and draw up the contents.
- For IV administration:
  - Dilute to approximately 10 ml using 0.9% sodium chloride and administer over 1-2 minutes.
- For IM administration:
  - Do not further dilute after reconstitution. The preferred IM site is the lateral thigh. If this site is not suitable use the lateral upper arm.

Common adverse effects

- Fatigue.
- Sodium and water retention. This may worsen hypertension and heart failure, but is usually only of clinical significance with repeated and/or prolonged dosing.

Usual onset of effect

- 10-20 minutes.

Usual duration of effect

- 8-12 hours.

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Usual preparation

- Vial with two chambers, one containing 100 mg of hydrocortisone powder and the other containing 2 ml of sterile water for reconstitution.

Pharmacokinetics

- Hydrocortisone is metabolised in the liver.
- Clearance is prolonged if the patient has significant liver impairment, but this does not alter the initial (loading) dose.

Common interactions

- None.

More Information

Version history

16 June 2026 1.1.0.1
Update: Indications updated.

13 February 2026 1.0.7.1
Correction: Removed advice of shaking vial until dissolved.

9 June 2025 1.0.5.3
Update: Clinical level, indications, dosage and administration information updated.